

## PUMaC 2015 Power Round

“We have a new theorem—that mathematicians can prove only trivial theorems, because every theorem that’s proved is trivial.” —Richard Feynman

### Rules and Reminders

These rules supersede any rules appearing elsewhere about the Power Round:

1. Your solutions are to be turned in when your team checks in on the morning of PUMaC or emailed to us at [pumac@math.princeton.edu](mailto:pumac@math.princeton.edu) by 8AM Eastern Standard Time on the morning of PUMaC, November 21, 2015 with the subject line “PUMaC 2015 Power Round.” Please staple your solutions together, including the Power Round cover sheet (the last page of this document) as the first page. Each page should also have on it the **team number** (not team name) and **problem number**. Solutions to problems may span multiple pages, but staple them in continuing order of proof.
2. It is not necessary to do the problems in order, although it is a good idea to read all the problems, so that you know what is permissible to assume when doing each problem. However, please collate the solutions in order in your solution packet. Each problem should start on a new page, and solutions should be written on one side of the paper only (there is a point deduction for not following this formatting).
3. On any **Problem**, you may use without proof any **Problem**, **Lemma**, or **Proposition** from earlier in the test, even if it’s a problem your team has not solved. You may cite results from conjectures or subsequent problems only if your team solved them independently of the problem in which you wish to cite them. You may not cite parts of your proof of other problems: if you wish to use a lemma in multiple problems, please reproduce it in each one.
4. All problems are numbered as “Problem x.y” where x is the section number. A list of problems follows the table of contents to help you keep track of the problems you have done. Each problem’s title and point distribution can be found in parentheses between the problem number and problem statement. The point distribution is bolded.
5. Using computer programs, calculators, and Mathematica (or similar programs) is allowed. However, **print and online references are not allowed**. If you attempt to use a program, please attach any relevant code that is used in the solution.
6. Teams whose members use English as a foreign language may use dictionaries for reference.
7. No communication with humans outside your team of 8 students about the content of these problems is allowed. If you have any questions regarding the test, please contact us at once at [pumac@math.princeton.edu](mailto:pumac@math.princeton.edu).

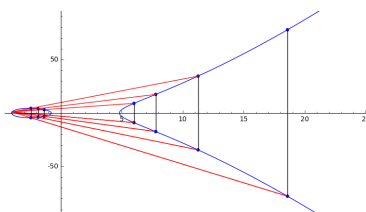


Figure 1: Graphical View of kP.

# 1 Introduction

Elliptic curves appear often in mathematics because they possess remarkably nice properties. For example, elliptic curves relate elegantly to ideas from Galois theory. This Power Round will demonstrate the utility of elliptic curves and Galois theory by using them to prove an interesting fact about the Tiger sequence. We define the Tiger sequence  $\{a_n\}$  for non-negative  $n$  recursively by  $a_0 = a_1 = a_2 = a_3 = 1$  and for  $n \geq 4$ , the relation

$$a_n a_{n-4} = a_{n-1} a_{n-3} + a_{n-2}^2.$$

We ultimately seek to prove the following theorem.

**Theorem 1.** *Let  $\pi(x)$  be the number of primes less than or equal to  $x$ . Then*

$$\lim_{x \rightarrow \infty} \frac{|\{p \leq x : p \text{ prime and } p|a_n \text{ for some } n\}|}{\pi(x)} = \frac{11}{21}.$$

This  $\frac{11}{21}$  fraction is called the density of primes that divides a term of the Tiger sequence.

While much mathematical machinery is needed to prove this, we have broken down the task into a series of sections that culminate in a final proof in section 9. Please note that while the ultimate goal of this Power Round is to prove the given theorem, the sections may include problems that are not essential to the final proof, but are relevant and good problems to try. We have sorted the problems in as straightforward a manner as possible with regards to the final proof, but as the various topics are very interconnected you may find it useful to refer back to previous sections for ideas on how to proceed. As always, refer to the rules at the start of the document for how to reference other problems.

In a similar vein, we have a couple housekeeping remarks:

- All definitions, propositions, lemmas, and theorems are labeled in increasing order using the same index. For example, this document began by introducing a theorem labeled Theorem 1 and will soon introduce its first lemma labeled Lemma 2 followed by its first official definition labeled Definition 3.
- While this document guides you through the final proof, it will not babysit your progress. In any given part of the document, we may make assertions that will be necessary when solving a later problem. It is your responsibility as the reader to keep track of such material. Details that are absolutely essential will often be written in bold, but this is not an if and only if criterion for discerning important facts.

Lastly, you may be asking yourself: “Why is this interesting?” Well, we could name-drop famous mathematicians who have answered similar questions, or lie and say this is a large area of mathematics (this specific flavor of math isn’t). But that would only tell you why this topic is interesting to others. Why will it be interesting to you? Well, if we take a slightly more general view, the question becomes: Why is math interesting to you? You have probably heard people say that math is necessary for science or as a life skill. But, if you have voluntarily decided to take this contest, you might have a different opinion. Sure, the science/life skills part might be true, but that’s not why you are here and that’s not why we have organized this tournament for you. PUMaC is a competition for many different types of people created by many different types of people, but we all share an interest in and appreciation for math for its own sake. Math is pretty. Back to the original question, we hope that this Power Round will expose you to an area of math you haven’t seen before and that is remarkably pretty (but we’ll leave that aesthetic judgment to you). Now, please don’t get too carried away by the scoring and the fact that this is a competition. It may be cliché, but please have fun as well.

-Heesu Hwang.

We’d like to acknowledge and thank many individuals and organizations for their support; without their help, this Power Round (and the entire competition) could not exist. Please refer to the solution of the Power Round for full acknowledgments.

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## 2 Let's Get Started

Firstly, if you haven't read the introduction, then go back and read the introduction! It really does add big-picture context to the problems you're doing. Now as a preliminary and as an example of the standards of justification we expect throughout this power round, here is a classical number theory result called Bézout's Lemma, with proof. Very quickly, here is a spot of terminology:

**Definition 2.**  $\mathbb{Z}$  represents the set of integers.  $\mathbb{Q}$  represents the set of rational numbers.  $\mathbb{R}$  represents the set of real numbers.  $\mathbb{C}$  represents the set of complex numbers. Finally,  $\mathbb{N}$  represents the set of positive integers. (A point of interest, Europeans include 0 in this set while Americans do not. Do you agree? Is 0 a natural number?)

**Definition 3.** *Sets.* A set is a collection of objects. We write that set  $A = \{a_i\}$  for integers  $1 \leq i \leq n$  if  $A$  contains exactly elements  $a_i$  and nothing else. We may similarly have infinite sets. Here is some specific notation with sets:

- $a \in A$ : Let  $A$  be a set such that  $A$  contains the element  $A$ ; then we write  $a \in A$ .
- $A \setminus B$ : Let  $A$  and  $B$  both be sets. Then  $A \setminus B$  denotes the set of elements that are inside  $A$  but not in  $B$ .
- $A \subset B$ : Let  $A$  and  $B$  both be sets. If every element of  $A$  lies inside  $B$ , we write  $A \subset B$ .
- $\forall a \in A$ : The notation  $\forall a \in A$  is read in English as “for all elements  $a$  in  $A$ ”, and means we consider all elements of set  $A$ .
- $\exists a \in A$ : This notation is read as “there exists an element  $a$  (in set  $A$ ).” It is usually followed up by the phrase “such that.” For example,  $\exists x \in \mathbb{R}$  such that  $x > 0$ .

**Lemma 4** (Bézout's Lemma). *Prove that if  $x$  and  $y$  are coprime integers, then there exist integers  $a$  and  $b$  such that  $ax + by = 1$ .*

*Proof.* We prove the more general case of integers  $x$  and  $y$  with a general gcd.

Examine the set  $S := \{c \in \mathbb{N} : c = ax + by \text{ where } a, b \in \mathbb{Z}\}$ . Since  $\gcd(x, y)$  divides  $x$  and  $y$ ,  $\gcd(x, y)$  divides any element of  $S$ .

Suppose that  $l$  is the least element of  $S$  by the well-ordering principle (which states that every non-empty set of positive integers has a least element); thus let  $a_0$  and  $b_0$  be integers such that  $l = a_0x + b_0y$ . Take any other arbitrary element of  $S$ ; for example  $k = a_1x + b_1y$ . Then by integer division, suppose that  $k = l \cdot q + r$  where  $0 \leq r < l$ . Thus

$$l = a_0x + b_0y \implies lq = a_0qx + b_0qy \implies r = k - lq = (a_1 - a_0q)x + (b_1 - b_0q)y.$$

Since  $l$  by assumption was the least positive integer that was a linear combination of  $x$  and  $y$ , we must have  $r = 0$ . Thus  $k = l \cdot q$ , and every element of  $S$  is divisible by  $l$ .

Note that  $|x|, |y| \in S$ ; thus  $l \mid \gcd(x, y)$ . But clearly  $\gcd(x, y) \mid l$ , and so they are equal. Thus there is some integer solution to  $l = ax + by$ .  $\square$

### 3 Group Theory

Groups are fundamental to mathematics. They form the basis (pun!) of algebra, one of the two overarching subjects of math. It is important that anyone who wishes to explore math further understands groups well. A group is defined as follows.

**Definition 5.** A group is a set of elements  $G$  with a closed binary operation  $*$ . Binary means  $*$  operates on two elements. Closed means that for any  $a, b \in G$ ,  $a * b$  is contained inside  $G$ . These elements and operation obey the following three rules.

- There exists an unique element  $e \in G$  such that for all elements  $g \in G$ ,  $e * g = g * e = g$ . This is called the identity element. (In cases of ambiguity, we will denote this as  $e_G$ .)
- For any element  $g \in G$ , there exists a unique element  $h \in G$  such that  $g * h = h * g = e$ . This element  $h$  is usually denoted  $g^{-1}$  (in additive groups (to be explained), it is denoted  $-g$ ).
- For all  $a, b, c \in G$ , the associative property holds:

$$a * (b * c) = (a * b) * c.$$

**Definition 6.** Suppose  $\{G, *\}$  is a group. Suppose  $*$  is commutative. That is, for all  $a$  and  $b$  in the group,  $a * b = b * a$ . Then  $G$  is called commutative or abelian.

To demonstrate all this,  $\mathbb{Z}$  is a group with addition as the operation. This is a group because we may always add two integers and get another integer, 0 is the additive identity, and similarly the other group properties are satisfied. In fact,  $\{\mathbb{Z}, +\}$  is an abelian group. Now here are some warm-up problems.

**Problem 3.1** (Basic Group Theory; **2, 2, 2, 2**). Give justification for each of the following.

- Is  $\{\mathbb{Z}, -\}$ , the set of integers under subtraction, a group?
- Is  $\{\mathbb{N}, +\}$ , the set of positive integers under addition, a group?
- Is  $\{\mathbb{Q}, \cdot\}$ , the set of rationals under multiplication, a group?
- Is  $\{\mathbb{Q} \setminus \{0\}, \cdot\}$ , the set of rationals without zero under multiplication, a group?

We turn now briefly to the topic of modular numbers where we find groups arising naturally. We denote  $\mathbb{Z}/n\mathbb{Z}$  as the integers mod  $n$ . Elements of this set are integers  $m$  where  $0 \leq m < n$ . Note that by division, for any integer  $z \in \mathbb{Z}$ , we may write  $z = n \cdot q + r$  where  $0 \leq r < n$  and  $q$  and  $r$  are both integers. Thus in this set of integers mod  $n$ ,  $z \equiv r \pmod{n}$ . For examples, in the set of integers mod 7 ( $\mathbb{Z}/7\mathbb{Z}$ ),  $10 \equiv 3 \pmod{7}$ ,  $-5 \equiv 4 \pmod{9}$ , and  $1063 \equiv 3 \pmod{10}$ . A nice property of mods is that we may substitute equivalent quantities at any time. For example,

$$\begin{aligned} 6546 \cdot 773 - 1650 \cdot 945 + 8^{654651} &= (935 \cdot 7 + 1) \cdot (110 \cdot 7 + 3) - (235 \cdot 7 + 5) \cdot (135 \cdot 7) + (7 + 1)^{654651} \\ &\equiv 1 \cdot 3 - 5 \cdot 0 + 1^{654651} \pmod{7} \\ &\equiv 4 \pmod{7}. \end{aligned}$$

We mention mods because they are very fundamental to number theory and algebra. For one, note that  $\mathbb{Z}/n\mathbb{Z}$  is a group with respect to addition (morally, you should really prove this to yourself, but there is no

problem to write up). If  $n = p$  is prime, we equivalently write  $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ . Primes are really nice numbers, and in general “things” like  $\mathbb{F}_p$  that are associated with primes are also often very nice, as we shall see in depth later (fields!). For now, here’s just a small problem about mods.

**Problem 3.2** (Finite Field; 5). *Let  $p \in \mathbb{Z}$  be a prime. Setwise, the unit group of  $\mathbb{F}_p$  is  $\mathbb{Z}/p\mathbb{Z} \setminus \{0\}$ . Prove that the unit group of  $\mathbb{Z}/p\mathbb{Z}$ , denoted as  $\mathbb{Z}/p\mathbb{Z}^\times$ , is a group under multiplication.*

Turning back now to group theory, if the operation of a group is called “addition,” we call it an additive group. Notationally, for any  $g \in G$  and  $n$  a non-negative integer,  $ng := \sum_{i=1}^n g$  denotes  $g$  added to itself  $n$  times and for  $n < 0$ ,  $ng = -((-n)g)$ . Also,  $0$  denotes the group identity. Similarly, if the operation is called multiplication, then for any  $g \in G$  and  $n$  a non-negative number,  $g^n = \prod_{i=1}^n g$  denotes  $g$  multiplied by itself  $n$  times, and for  $n < 0$ ,  $ng = (|n|g)^{-1}$ . Also,  $1$  is the group identity. When notation is written with powers, it is implicitly implied that the operation is multiplication, and similarly notation that uses scalar multiples of group elements implies the operation is addition. With that, here are a bunch of problems and definitions!

**Definition 7.** *The order of any element  $g \in G$  is the least positive integer  $l$  such that  $g^l = e$ . If no such  $l$  exists, then the order is called infinite.*

*Secondly, the order of a group  $G$  is the number of elements of  $G$ . Note if  $G$  has infinitely many elements, the order of  $G$  is infinity.*

For example, 2015 has infinite order in the group of integers under addition. However, 2015 has order 2 in  $\mathbb{Z}/4030\mathbb{Z}$ .

**Problem 3.3** (Finite Group; 5). *Let  $G$  be a finite group. Prove that every element of  $G$  has a finite order.*

We only consider finite groups in depth during this power round. However, we have seen examples of infinite groups (such as the integers under addition). In general, there aren’t really large philosophical differences between the two categories. However, we primarily present finite groups here because it’s easier to work with them when first starting out.

**Definition 8.** *Suppose that  $G$  is a finite group and  $H \subset G$  is a non-empty subset of the elements of  $G$  that is itself a group. Then  $H$  is called a subgroup of  $G$  where  $H$  inherits the operation of  $G$ .*

The motivation behind such a construction of a subgroup is simple enough; it is a group in its own right that just happens to be contained inside another! You know what, here’s a fun trick that you may find useful sometime (you likely won’t use it in this power round, unfortunately).

**Problem 3.4** (Subgroup Test; 5). *Let  $G$  be a group and  $H \subset G$  a non-empty subset of  $G$ . Suppose  $H$  is a set that has the property that for all  $a, b \in H$ ,  $ab^{-1} \in H$ . Prove that  $H$  is a subgroup of  $G$ .*

Subgroups in fact are very nice, natural constructions from groups. Think back to the group of integers and the group of integers mod  $n$ ; for example let’s take  $n = 7$ . Another way to think about mods is to imagine that the numbers  $0 \leq k < 7$  simply represent the classes of numbers  $\{c \cdot 7 + k : c \in \mathbb{Z}\}$ .

For example, 0 represents the class of numbers  $[0] := \{\dots, -14, -7, 0, 7, 14, \dots\}$ , 1 represents the class  $[1] := \{\dots, -13, -6, 1, 8, 15, \dots\}$ , and so forth. In this manner, note that  $[0]$  is a subgroup of  $\mathbb{Z}$  under addition. However,  $[1]$  is unfortunately not a subgroup. For example,  $1 + 1 = 2$  is not in the set  $[1]$ . However,  $[1]$  is still a pretty natural construction; maybe there is a name for it? Yes, there is!

**Definition 9.** Let  $H$  be a subgroup of  $G$ . For all  $g \in G$ , we call  $gH := \{g \cdot h : h \in H\}$  a left coset. For all  $g \in G$ , we call  $Hg := \{h \cdot g : h \in H\}$  a right coset.

For example, let  $H = 7\mathbb{Z}$ , the set of numbers that is 7 times an integer, and let  $G = \mathbb{Z}$  the integers. Note that  $H$  is  $[0]$  as mentioned above. Then  $1 + H$  is a coset of  $G$  (note in this case, this is both a left and right coset) that in this case is the class  $[1]$  mentioned directly above.

Cosets are very valuable tools in the proofs of group theory problems. Why? Because they are very symmetric in a sense, and this leads to neat properties. For example, if  $aH$  and  $bH$  are two cosets of a group  $G$ , do you think they have to intersect non-trivially? Well no;  $[0]$  and  $[1]$  as we saw above do not intersect ( $[0]$  is the coset  $0 + H$  in our example). Well if  $aH$  and  $bH$  do intersect, do you think they can intersect only a little bit? That is, can  $aH \cap bH$  be non-empty, but  $aH \cap bH$  is strictly smaller than both  $aH$  and  $bH$ ? Or if  $G$  is a finite group, do you think that  $aH$  and  $bH$  have to have the same size? Or do you think that every element of  $G$  has to be contained inside some coset? These are all questions that you should think about.

Finally, this leads us to a what we call Lagrange's Theorem.

**Problem 3.5** (Lagrange's Theorem; 10). Suppose  $G$  is a finite group of order  $n$ . Then prove for all  $g \in G$ ,  $g^n = 1$ .

And now, before we proceed to other sections, here is a final group theory problem.

**Problem 3.6** (Odd Order; 5). Let  $G$  be a finite group. Prove that an element  $g$  has odd order in  $G$  if and only if for every positive integer  $k$ , there exists some element  $h_k \in G$  such that  $(h_k)^{2^k} = g$ . (Note that we do not specify  $G$  as abelian; it is sufficient to take  $G$  a finite group.)

Hopefully you have found these problems illuminating (hint: you may see these things again). However, to end this section, we would just like to present some standard material that we think is enriching to experience.

Suppose we have two groups  $G$  and  $H$ , and we create a function  $f$  between them  $f : G \rightarrow H$ . But we don't want any old functions; we want functions that somehow show that the structure of  $G$  and  $H$  have similarities. For example, examine  $\mathbb{Z}$  and  $2\mathbb{Z}$  as additive groups ( $2\mathbb{Z}$  is notation for the set of all even numbers). Any addition done with the even numbers is linked to addition done with regular numbers. For example,

$$2 + 46 = 48 \Leftrightarrow 2(1) + 2(23) = 2(24).$$

We want functions between groups to somehow reflect the similarities between them. Thus we impose more conditions on  $f$ .

**Definition 10.** Let  $G$  and  $H$  be groups with a function  $f : G \rightarrow H$ . This function  $f$  is called a homomorphism if:

- For all  $a, b \in G$ ,  $f(a *_G b) = f(a) *_H f(b)$  ( $*_G$  and  $*_H$  represent the operations of  $G$  and  $H$  respectively).

For example, let  $G = \mathbb{Z}$ ,  $H = 2\mathbb{Z}$ , and let  $f(a) = 2a$ . We encourage you to check that this is indeed a homomorphism, and that it has the property we wanted: that it represents the relationship between the structure of  $G$  and  $H$ .

**Definition 11.** Let  $f : G \rightarrow H$  be a homomorphism. If every element of  $H$  is the image of some element of  $G$ , then  $f$  is surjective and is a surjection.

**Definition 12.** Let  $f : G \rightarrow H$  be a homomorphism. If the only element of  $G$  that is mapped to  $e_H$ , the identity element of  $H$ , is  $e_G$ , then  $f$  is injective and is an injection.

**Definition 13.** If a group homomorphism is both injective and surjective (if it satisfies both, then it is also called bijective), then it is called an isomorphism.

You may want to convince yourself that in the example above, the homomorphism is indeed bijective. In general, the same terminology applies to general functions. If a function  $f : A \rightarrow B$  has the property that  $\forall b \in B, \exists a \in A$  such that  $f(a) = b$  (surjective) and that  $\forall a, a' \in A, f(a) = f(a') \implies a = a'$  (injective), then  $f$  is bijective (this is an adjective) and is a bijection (this is a noun).

Next, we introduce the idea of mashing groups together. Recall that a number line can be represented as  $\mathbb{R}$ . A coordinate plane is similarly represented by the notation  $\mathbb{R}^2$  or  $\mathbb{R} \times \mathbb{R}$ . This is called a direct product, and elements of  $\mathbb{R}^2$  are coordinate pairs  $(a, b)$  where  $a$  and  $b$  both come from  $\mathbb{R}$ . Well, this is something we can do with groups as well. For example, taking two groups  $G$  and  $H$ , we denote  $G \times H := \{(g, h) : g \in G, h \in H\}$ . The group operation on this new set is the combination of the operations of  $G$  and  $H$  component-wise respectively. One fact that we ask the reader to convince themselves (this is clearly grammatically not “correct,” but the author strongly believes this should be an official singular, gender-neutral pronoun) is that  $G \times H$  is itself a group. While this isn’t an official problem, you should convince yourself that this works because it’s necessary to understand for the next part.

We first introduce a definition. We will explore this topic more thoroughly in section 7, but the definition is sufficient here for now.

**Definition 14.** Let  $R$  be a set of elements with a closed binary operation we call addition such that  $\{R, +\}$  is an additive, commutative group. Furthermore, suppose  $R$  admits another closed binary operation  $\cdot$  that we can call multiplication; it is commutative, and  $\forall a, b, c \in R, a \cdot (b \cdot c) = (a \cdot b) \cdot c$ ;  $a \cdot (b + c) = a \cdot b + a \cdot c$ ; and  $(b + c) \cdot a = b \cdot a + c \cdot a$ . Finally, there exists a multiplicative identity we call 1 such that for all  $a \in R, a \cdot 1 = 1 \cdot a = a$ . Then  $R$  is called a ring.

**Definition 15.** Let  $R$  be a ring. Then  $GL_n(R)$  is called the general linear group, and denotes the group of  $n \times n$  invertible matrices with elements in  $R$  as a group under multiplication.

There is yet another way to make a group from two smaller groups. As a generalization of the direct product, we introduce the semidirect product. We start with a group  $G$  and a group  $H$ , where  $H$  is a group of functions that act on elements of  $G$ . For example, let  $G$  be the set of vectors  $\mathbb{Z} \times \mathbb{Z}$ , and let  $H$  be the set of matrices  $GL_2(\mathbb{Z})$ .

In other words,  $G$  is a set of vectors of the form  $(a, b)$  where  $a$  and  $b$  are both integers, and  $H$  is the set of invertible  $2 \times 2$  matrices with integer coefficients (invertible under multiplication). Elements of  $H$  are indeed functions that act on elements of  $G$ ; for example if  $h \in H$  and  $g \in G$ , then  $h \cdot g$  is another vector. Thus we can define a semidirect product as follows.



**Proposition 16.** Let the semidirect product setwise be defined as  $G \rtimes H := \{(g, h) : g \in G, h \in H\}$  where  $H$  is a group of functions that acts on  $G$  (yes, groups may have functions as elements). Let  $*_g$  and  $*_h$  denote the group operations of  $G$  and  $H$  respectively. Then the group operation on this set for  $(g_1, h_1), (g_2, h_2) \in G \rtimes H$  is defined by

$$(g_1, h_1) * (g_2, h_2) := (g_1 *_g h_1(g_2), h_1 *_h h_2).$$

$G \rtimes H$  is a group.

Thus in our example here, we can define a group  $(\mathbb{Z} \times \mathbb{Z}) \rtimes \text{GL}_2(\mathbb{Z})$  (this is an example of an affine general linear group as we shall see later). Here is an example operation:

$$\begin{aligned} \left( \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 & 2 \\ 2 & -1 \end{bmatrix} \right) * \left( \begin{bmatrix} 1 \\ 5 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) &= \left( \begin{bmatrix} 2 \\ 0 \end{bmatrix} + \begin{bmatrix} -5 & 2 \\ 2 & -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 5 \end{bmatrix}, \begin{bmatrix} -5 & 2 \\ 2 & -1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) \\ &= \left( \begin{bmatrix} 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 5 \\ -3 \end{bmatrix}, \begin{bmatrix} -5 & 2 \\ 2 & -1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) \\ &= \left( \begin{bmatrix} 7 \\ -3 \end{bmatrix}, \begin{bmatrix} -5 & 2 \\ 2 & -1 \end{bmatrix} \right). \end{aligned}$$

Later on when we revisit this affine general linear group, we use this notation  $AGL_2(R)$  to denote  $(R)^2 \rtimes \text{GL}_2(R)$  where  $R$  is a general ring. We will go in depth about this later. With this, we turn now to the topic of elliptic curves.

## 4 Elliptic Curves

Elliptic curves are integral (hah! it's a pun!) to mathematics, and in fact have even higher generalizations called varieties. For the purposes of this power round, we define an elliptic curve as follows.

**Definition 17.** An elliptic curve  $E$  is the curve satisfying an equation of the form

$$E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6,$$

where the coefficients  $a_i$  are INTEGERS (can't stress this enough :p).

The strange coefficient numberings are of historical significance. The reason that elliptic curves are interesting is that there is a natural group on the set of points on them.

**Very important remarks:** for the entirety of this power round, we take coefficients of elliptic curves to be integers. However, we may allow points on the curve to be rational or something else. Unless otherwise specified, any elliptic curve has integer coefficients and any points we consider on it are rational points.

Now, if you're the average person looking at this equation, you may be slightly disgusted by how unwieldy it looks; come on, that  $xy$  term looks atrocious. So let's get rid of it.

**Problem 4.1** (Transformation of EC; **2, 8**).

- Let  $f(x) = x^3 + a_2x^2 + a_1x + a_0$  be a polynomial with rational coefficients. Find some linear change of variables  $x = mx' + n$  such that  $f(x) = x'^3 + b_1x' + b_0$  is another polynomial with rational coefficients.
- Let  $E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$  be an elliptic curve. Find a change of variables  $y \mapsto y'$  and  $x \mapsto x'$  such that  $y'^2 = x'^3 + Ax' + B$ . This gives us another elliptic curve  $E'$ ; note that  $E'$  is an elliptic curve, and so  $A, B \in \mathbb{Z}$ . (Hint: you may have to use a change of variables that involves two variables at once. Also keep in mind that the starting and final coefficients have to be integral.)

This latter form is what is called the short Weierstrass form. This form can be much easier to work with at times. Substitution for  $y^2$  is a lot easier for example. Let's try to work with this form.

Let  $E : y^2 = x^3 - 20x - 15$  be an elliptic curve. Note the points  $P = (-4, 1)$  and  $Q = (-1, -2)$  lie on  $E$ . The picture below shows the graph of the elliptic curve in blue with two points on it  $P$  and  $Q$ . In the picture, notice the black line through points  $P$  and  $Q$ . It intersects the elliptic curve again at another point we call  $P * Q$  at coordinates  $(6, -9)$ . Finally, the reflection of point  $P * Q$  vertically over the  $x$ -axis is shown by the red line. This gives us a point we call  $P + Q$  at  $(6, 9)$ . Thus we write  $(-1, -2) + (-4, 1) = (6, 9)$ .

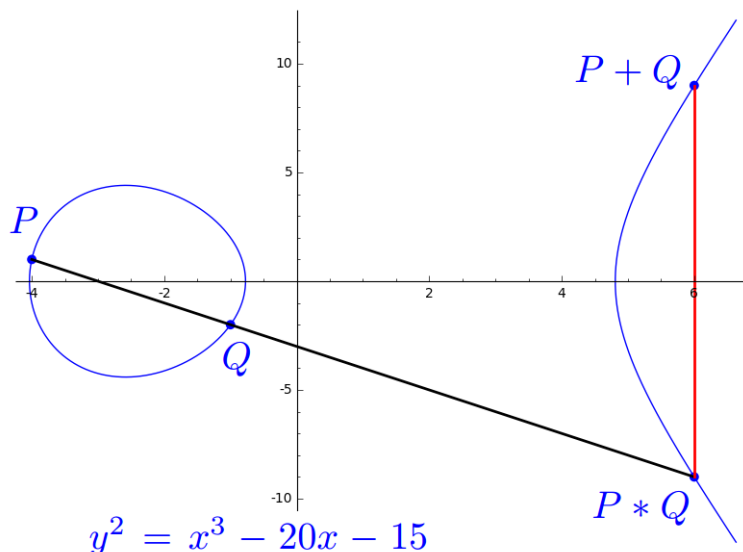


Figure 2: Addition of Two Rational Points.

This method in general gives us a way to define the “addition” of two points that will lead us to a group. Let's do a few more examples first, though. We stress that getting  $P + Q$  from  $P * Q$  by reflecting over the  $x$ -axis only works for elliptic curves in short Weierstrass form.

**Problem 4.2** (Addition Computation; **2, 2**). Let  $E : y^2 = x^3 - 20x - 15$  be an elliptic curve. Note that  $(6, -9)$ ,  $(-4, 1)$ ,  $(6, 9)$ , and  $(204, 2913)$  all lie on the curve.

- What is  $(6, -9) + (-4, 1)$ ?
- What is  $(6, 9) + (204, 2913)$ ?

We are ready to present the group on an elliptic curve. Again, we reiterate that we have only seen addition for elliptic curves in short Weierstrass form.

**Definition 18.** Let  $E$  be a general elliptic curve. Then  $E(\mathbb{Q})$  denotes the set of rational points on  $E$ . That is, those points  $(\alpha, \beta)$  with  $\alpha, \beta \in \mathbb{Q}$  such that  $(\alpha, \beta)$  lies on the curve  $E$ .

**Definition 19.** Let  $E$  be a general elliptic curve with points  $P$  and  $Q$  on  $E$ . Note that line  $PQ$  intersects  $E$  at a third point. This point is called  $P * Q$ . (The cautious reader might see some problems with this definition. For example, what if  $P = Q$ ? This statement will be fully justified later, although you may be able to prove it yourself.)

**Proposition 20.** Let  $E$  be an elliptic curve in short Weierstrass form. Then  $\{E(\mathbb{Q}), +\}$  is a group where the binary operation on two points  $P$  and  $Q$  is called addition. Namely,  $P + Q$  is the reflection of  $P * Q$  over the  $x$ -axis.

Proving that this group (incredibly) exists can be unnecessarily complicated in general. For example, the existence of the identity element requires machinery that is a little too complicated to build here (we will build it later though!). Also, digest for a moment the almost magical nature of what this says. Given a line through two rational points on an elliptic curve, you get another rational point! With this in mind, it is a good exercise to prove some of the basic facts of this group law, after which you may assume this proposition is true.

**Problem 4.3** (Addition Theory; **2, 8**). Let  $E : y^2 = x^3 + Ax + B$  be an elliptic curve with  $A, B \in \mathbb{Z}$  and  $P = (a, b)$ ,  $Q = (c, d)$ , and  $H = (e, f)$  be rational points on  $E$ . For simplicity, we assume the  $x$ -coordinates of  $P$ ,  $Q$ , and  $H$  are distinct.

- a) Prove that  $P + Q$  is a rational point.
- b) Prove that the associative property holds. Namely, show that  $(P + Q) + H = P + (Q + H)$ . You may furthermore assume the  $x$  coordinates of  $P + Q$  and  $H$  are different, and that of  $P$  and  $Q + H$  are different as well.

**Problem 4.4** (Reduced Rational Point; **5**). Let  $E : y^2 + e_1xy + e_3y = x^3 + e_2x^2 + e_4x + e_6$  be an elliptic curve with integer coefficients. Suppose that  $P = \left(\frac{a}{b}, \frac{c}{d}\right)$  on  $E$  is a rational point in reduced form (i.e. both the coordinates are reduced fractions). We may assume  $b$  and  $d$  are positive (since  $a$  or  $c$  can be negative). Give an equality relating  $b$  and  $d$  by writing one as a positive power of the other.

So if you are morally convinced by now that  $E(\mathbb{Q})$  should be a group (as you should be), then you might also morally accept that Proposition 20 is also true for any elliptic curve, not just those written in short Weierstrass form. And it is! However, the definition of addition is slightly different in the general case. The reason we took  $P + Q$  as the reflection of  $P * Q$  in the short Weierstrass case is that for elliptic curves written in short Weierstrass form, there is a natural horizontal line of symmetry at the  $x$ -axis. In the general case, we can find another natural horizontal line of symmetry; and so in the general case, while  $P * Q$  is always the same,  $P + Q$  will be different.

For example, consider the curve  $y^2 - 2015y = x^3 - 36x^2 + x$ . Notice that for all points  $(a, b)$  on the curve, by how left hand side is written, the point  $(a, 2015 - b)$  is also on the curve. Thus the horizontal line of symmetry in this case is  $y = \frac{2015}{2}$ . For this curve then,  $P * Q$  is defined as the third point of intersection of the line  $PQ$  and the curve, and  $P + Q$  is defined as the reflection of  $P * Q$  over the line  $y = \frac{2015}{2}$ . More generally for all the curves we consider in this power round, this process of reflecting over the line of symmetry applies.

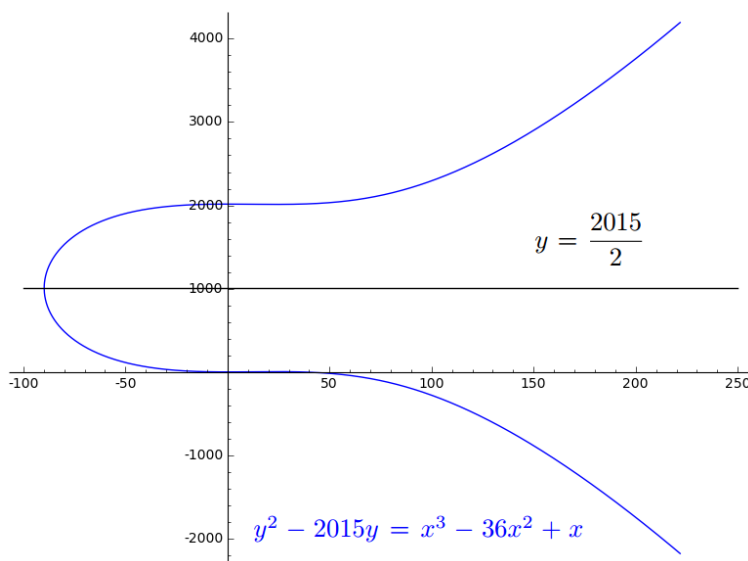


Figure 3: Horizontal Line of Symmetry.

**Definition 21.** Let  $E$  be an elliptic curve. Then  $P + Q$  is the reflection of  $P * Q$  over the horizontal line of symmetry of  $E$ .

Consider now the elliptic curve  $E : y^2 + y = x^3 - x$  and the point  $P = (0, 0)$  on this curve. From this point on, we mostly refer to this curve and point  $P$ . It is the most important curve that we examine in order answer our question about prime density.

**Problem 4.5** (E Symmetry; **2, 2**). Let  $E$  be the elliptic curve  $y^2 + y = x^3 - x$ .

- For every point  $(a, b)$  on  $E$ , there is another point  $(a, c)$  on  $E$  as well. What is  $c$ ?
- There is the horizontal line of symmetry  $y = \alpha$  for this curve  $E$ . What is  $\alpha$ ?

Now, remember that sequence we mentioned in the introduction? Recall that the Tiger sequence  $\{a_n\}$  is defined by  $a_0 = a_1 = a_2 = a_3 = 1$  and recursively by  $a_n a_{n-4} = a_{n-1} a_{n-3} + a_{n-2}^2$ . You'll get to prove some facts about this sequence later. This following problem should keep you occupied with it until then. Keep in mind that we fixed a curve  $E$  and point  $P$  above. One fact about  $E$  and  $P$  is that  $P + P = (1, 0)$  (for the interested, adding a point to itself requires the use of a tangent line; learn calculus!).

**Problem 4.6** (Sequence and Curve; **10**). Prove for  $n > 1$  that  $(2n - 3)P = \left(\frac{f(n)}{a_n^2}, \frac{g(n)}{a_n^3}\right)$  where  $f(n) = a_n^2 - a_{n-1}a_{n+1}$  and  $g(n) = a_{n-1}^2 a_{n+2} - 2a_{n-1}a_n a_{n+1}$ . (Recall from section 3, the group theory section, that  $kP = \sum_{i=1}^k P$ .)

## 5 Sequences

So we turn now to the Tiger sequence again. A priori, we know nothing about this sequence. From the definition, it's not even clear that it's integral! (Hint: it is). As reference, here are some of the first few values starting with  $a_0$ : 1, 1, 1, 1, 2, 3, 7, 23, 59, 314,  $\dots$ . Well, that is royally unhelpful. Let's try to get our hands dirty working with these types of non-linear recurrences. Note by the recursive definition of the Tiger sequence that we may define terms of the sequence for negative  $n$ . For example,  $a_{-1}a_3 = a_0a_2 + a_1^2$  gives a way to define  $a_{-1}$ . This may be necessary for you to establish base cases.

**Definition 22.** For  $n \in \mathbb{N}$ , define  $s_n := a_{n-3}a_{n+3} - a_{n-2}a_{n+2}$ .

As it turns out, this new sequence of numbers is intimately related to the Tiger sequence, which may help us prove integrality.

**Problem 5.1** (Secondary Sequence; **5, 2**).

- a) Prove that  $a_n^2 s_{n-1} = a_{n-1}^2 s_n$  for  $n \in \mathbb{N}$ .
- b) Prove that  $s_n = 4a_n^2$  for  $n \in \mathbb{N}$ .

**Problem 5.2** (Is Integral; **10**). Prove that  $a_n$  is integral for  $n \in \mathbb{N}$  and that the following are true,  $\gcd(a_n, a_{n-1}) = \gcd(a_n, a_{n-2}) = 1$ .

Because there are so many ways to create a recursive sequence, there aren't really centralized strategies for dealing with them in much generality. But maybe this set of problems was interesting. As a parting shot, here are few more problems.

**Problem 5.3** (Sequence Divisibility; **5**). We define a recursive sequence  $\{b_n\}$  by  $b_0 = b_1 = b_2 = 1$  and for  $n \geq 3$ ,  $b_n = b_{n-1}b_{n-2} + b_{n-3}$ . Prove that for all integers  $n > 1$ , there exists a  $k \geq 0$  such that  $n|b_k$ .

**Problem 5.4** (Integrality, Integrality!; **8, 12**).

- a) We define a recursive sequence  $\{c_n\}$  by  $c_0 = c_1 = c_2 = c_3 = c_4 = 1$  and for  $n \geq 5$ ,  $c_n c_{n-5} = c_{n-4} c_{n-1} + c_{n-2} c_{n-3}$ . Prove that this sequence is integral for  $n \geq 0$ .
- b) We define a recursive sequence  $\{d_n\}$  by  $d_0 = 1$ ,  $d_1 = 2$ ,  $d_2 = 1$ , and  $d_3 = -3$  and for  $n \geq 4$ ,

$$d_n = \begin{cases} \frac{d_{n-1}d_{n-3} - d_{n-2}^2}{d_{n-4}} & \text{if } n \equiv 0, 1 \pmod{3} \\ \frac{d_{n-1}d_{n-3} - 3d_{n-2}^2}{d_{n-4}} & \text{if } n \equiv 2 \pmod{3}. \end{cases}$$

Prove that the sequence  $\{d_n\}$  is an integral sequence for  $n \geq 0$ .

## 6 Interlude

As the reader may have noticed by now, this Power Round is a rather eclectic collection of math topics. The following rephrases our previous work into a form we may use later.

We introduce projective space, specifically projective 2-space, denoted  $\mathbb{P}^2(\mathbb{R})$  (this means that coordinates are elements of  $\mathbb{R}$ ; we can also work instead in  $\mathbb{P}^2(\mathbb{Q})$ , but that is not necessary). The motivation of such a system of numbers is hard to flesh out fully here. (For the interested reader, consider this system as an attempt to fix the “problem” that two parallel lines do not intersect by adding points at infinity. For the artists out there, this is a formalization of the concept of perspective drawings in which parallel lines do in fact converge. Unfortunately, the implementation we present here may not make it clear why these things are true.)

Elements of  $\mathbb{P}^2(\mathbb{R})$  represent the lines in  $\mathbb{R}^3$ , real 3-space, that pass through the origin. Examine such a line  $\ell$  that passes through the origin  $(0, 0, 0)$ . We represent  $\ell$  by a triplet of coordinates  $(a : b : c)$  where  $\ell$  passes through points  $(0, 0, 0)$  and  $(a, b, c)$ . This clearly doesn’t give a unique representation of  $\ell$ . Under this representation, for all real numbers  $s$ ,  $(sa : sb : sc)$  and  $(a : b : c)$  will always represent the same line. For example, if  $\ell$  is a line that passes through  $(0, 0, 0)$  and  $(2, 4, 3)$ , then we can denote this line in  $\mathbb{P}^2(\mathbb{R})$  in many ways:  $(2 : 4 : 3) \cong (4 : 8 : 6) \cong (\pi : 2\pi : \frac{3\pi}{2})$ , and so forth. When possible, it is convention to standardize the way we represent these vectors by making the last coordinate 1; thus if  $c \neq 0$ , then  $(a : b : c) = (\frac{a}{c} : \frac{b}{c} : 1)$ , and the latter is the preferred form.

**Definition 23.** Let  $\mathbb{P}^2(\mathbb{R})$  represent projective 2-space. Then elements  $\alpha \in \mathbb{P}^2(\mathbb{R})$  are represented as  $\alpha = (a : b : c)$  where if  $c \neq 0$ , we may assume  $c = 1$ .

The reason we introduce this space is because it is in some sense the “correct” medium in which to examine elliptic curves.

**Definition 24.** Let  $E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$  be an elliptic curve. Denote another curve in three variables (adding  $z$ ) as  $F : y^2z + a_1xyz + a_3yz^2 = x^3 + a_2x^2z + a_4xz^2 + a_6z^3$ . We say that  $F$  has been homogenized because each term has the same degree (degree is the sum of the powers of all variables).  $F$  is an elliptic curve in  $\mathbb{P}^2(\mathbb{R})$ .

By the three variables given by homogenization, we can start to look at  $F$  as a curve lying in projective 2-space. It’s clear that if  $(a, b)$  is a solution to  $E$ , then  $(a : b : 1)$  is a solution to  $F$ . We may say more.

**Proposition 25.** There is a bijective correspondence between an elliptic curve  $E$  and a homogenized  $F$  with a further bijective correspondence between points on  $E$  and  $F$ .

We check this by proof by example! (Note this is not actually a proof. Never actually do this, but this example should illustrate clearly why this proposition is true.) Examine the elliptic curve  $E : y^2 = x^3 - 20x - 15$  again. Then the homogenization is  $F : y^2z = x^3 - 20xz^2 - 15z^3$ . Since  $(-4, 1)$  is a solution to  $E$ , we see  $(-4 : 1 : 1)$  is clearly a solution to  $F$ . Conversely note  $(1284 : -5601 : 64)$  is a solution to  $F$  (you may want to check this). Then it is clear  $(\frac{321}{16} : \frac{-5601}{64} : 1)$  is also a solution to  $F$ , and so  $(\frac{321}{16}, \frac{-5601}{64})$  is a solution to  $E$ . Note the importance of homogenization in this work. For example, what would have failed if  $F$  were made by simply multiplying every term by exactly one factor  $z$ ?

And finally, here is why we needed projective space: how do we look at elliptic curves over a finite field? An example of a finite field is  $\mathbb{F}_p$  (you know enough to verify that is indeed a field). Fields are explored more thoroughly in section 7.

**Definition 26.** Let  $K$  be a ring. Furthermore let  $K$  also have the additional property that for all non-zero elements  $a$ ,  $\exists a^{-1} \in K$  such that  $a \cdot a^{-1} = a^{-1} \cdot a = 1$ . Then  $K$  is called a field.

Perhaps it is unclear here why we would want to look at elliptic curves over  $\mathbb{F}_p$ , but you'll see why soon enough. So, is there a notion of an elliptic curve over  $\mathbb{F}_p$  where  $p$  is a prime? In case you haven't noticed by now how these rhetorical questions go... Yes! There is. Suppose  $E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$  is an elliptic curve with integer coefficients. We can look at points in  $E/\mathbb{F}_p$  (read  $E$  over  $\mathbb{F}_p$  meaning that we take  $E$  as an equation and points on  $E$  all inside  $\mathbb{F}_p$ ) in two ways: by solving  $E$  in  $\mathbb{F}_p$  from the start, or by reducing points from  $E/\mathbb{Q}$ . The first is easy to do: simply solve  $E : y^2 + a_1xy + a_3y \equiv x^3 + a_2x^2 + a_4x + a_6 \pmod{p}$ .

**Problem 6.1** (EC over Finite Field; 5). Let  $E : y^2 = x^3 + 3x + 9$  be an elliptic curve over  $\mathbb{F}_{13}$ . Find all the elements of  $E(\mathbb{F}_{13})$ . (Hint: there are 14 total elements. You may have to read on first to find the identity element.)

Otherwise, we can find points on  $E$  over finite fields by mapping rational points of  $E$  over  $\mathbb{Q}$  by the “obvious” mapping to try. Suppose that  $(\frac{a}{b}, \frac{c}{d}) \in E(\mathbb{Q})$  is a rational point on  $E$ . We look at the reduction of  $E$  onto  $\mathbb{F}_p$  by first translating to projective space; this point naturally maps to  $(ad : bc : bd)$ . Here in projective space, we divide by any powers of  $p$  necessary such that  $\gcd(ad/p^k, bc/p^k, bd/p^k) = 1$  (else the point would vanish trivially over  $\mathbb{F}_p$ ). Finally we translate into  $\mathbb{F}_p$  by taking these coordinates modulo  $p$ . Thus in summary,  $(\frac{a}{b}, \frac{c}{d}) \mapsto (ad \pmod{p}, bc \pmod{p}, bd \pmod{p})$ , modulo some conditions on clearing denominators with powers of  $p$ .

As promised before, we can now present the identity of the group  $E(\mathbb{Q})$ : it's  $(0 : 1 : 0)$ . This furthermore shows that the identity of  $E(\mathbb{F}_p)$  is also  $(0 : 1 : 0)$ . This gives us the following corollary (corollary of what I wonder...)

**Problem 6.2** (An Odd Divisor; 5). Let  $p$  be a prime. Prove  $p$  divides some term of the Tiger sequence  $\{a_n\}$  if and only if  $P = (0, 0)$  has odd order in the group  $E(\mathbb{F}_p)$  where  $E : y^2 + y = x^3 - x$ .

This is a rather magical connection between divisibility of a sequence and elliptic curves, don't you think? However, strangely enough, we will soon be able to make even weirder equivalent statements.

## 7 Galois Theory

Unfortunately, for the sake of time (we can't build up all of Galois theory from scratch for this Power Round!), we won't be able to give more than a heuristic of some of the methods we use here. (Un)luckily for you, the reader, this also means there aren't many problems directly on Galois theory :(. We hope that regardless of your mathematical background, this section is still interesting enough to try to understand. We describe first some of the necessary groundwork.

We previously introduced a fundamental object of algebra: groups. This was essentially the most basic “thing” we could do math on. We have only one operation on a group at all times. Anything simpler would have *very* little to it. Since then, we further saw a taste of more complicated algebraic objects, which as promised, we explore here. One step up from the group is another essential object of mathematics: a ring. Rings in some sense can be thought of as an extension of (additive) groups.

**Definition 27.** Let  $R$  be a set of elements that has a closed, binary operation we call addition such that  $\{R, +\}$  is an additive, commutative group and a second closed, binary, commutative operation  $\cdot$  that we can call multiplication. Suppose  $R$  has these properties:

- The operation  $\cdot$  is associative.

- For all  $a, b, c \in R$ ,

$$a \cdot (b + c) = a \cdot b + a \cdot c,$$

and

$$(b + c) \cdot a = b \cdot a + c \cdot a.$$

- Finally, there exists a multiplicative identity we call 1 such that for all  $a \in R$ ,  $a \cdot 1 = 1 \cdot a = a$ .

Then  $R$  is called a ring.

The consequences of such a definition is that  $R$  contains 0 (necessary by the addition law), and 1 (necessary by the multiplication law). Many definitions also add that  $1 \neq 0$ , but this is not strictly necessary.

**Problem 7.1** (0 Ring; 5). Let  $A := \{0\}$  be the set of just the element 0. Let  $+$  and  $\cdot$  be operations on  $A$  such that  $0 + 0 = 0 \cdot 0 = 0$ . Prove or disprove that  $A$  is a ring.

If you find the definition of rings a little scary looking, all it really says is that a ring is something like the integers  $\mathbb{Z}$ . (Mathematicians have this tendency of taking familiar objects like the integers and building abstractions of them. If you see one of these abstractions first, they can seem intimidating. But, if you know where the abstraction came from, you might see that it is quite natural. Rings are one example of this.) However, if you recall from the exercises in section 3, the integers are missing something. This leads us to something else called a field.

**Definition 28.** Let  $K$  be a ring where  $1 \neq 0$ . Furthermore let  $K$  also have the additional property that for all non-zero elements  $a$ ,  $\exists a^{-1} \in K$  such that  $a \cdot a^{-1} = a^{-1} \cdot a = 1$ . Then  $K$  is called a field.

Again, this is an example of an abstraction of a natural object. A field really emulates the properties of the rational numbers  $\mathbb{Q}$ . In the same way  $\mathbb{Q}$  is built from  $\mathbb{Z}$ , fields are built from rings. Also, note the relationship between rings and fields. A field is always a ring, but not the other way around.

There are many examples of fields. While  $\mathbb{Q}$  may have been a motivating example for the abstraction for a field (history fun fact: I have no idea if this is true. I made it up because this might be true...), fields are so common that you probably already know many other examples. A few more examples of fields are the real numbers  $\mathbb{R}$ ; the complex numbers  $\mathbb{C}$ ; and finite fields  $\mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p$ . Something of interest is that the



rational numbers are a subset of the real numbers; we write this  $\mathbb{Q} \subset \mathbb{R}$  and say that  $\mathbb{Q}$  is a sub-field of  $\mathbb{R}$ ; equivalently, we also say that  $\mathbb{R}$  is a field extension of  $\mathbb{Q}$ . Another example is  $\mathbb{R} \subset \mathbb{C}$ . We can say much more about the relationships between fields than characterizing them as subsets of each other.

The only thing that  $\mathbb{R}$  really lacks compared to  $\mathbb{C}$  is the number  $i$ . Every complex number is the combination of a real part and an imaginary part. This gives us another way to construct  $\mathbb{C}$ . We may write  $\mathbb{C} \cong \mathbb{R}[i]$ . This notation  $\mathbb{R}[i]$  means we take the set of real numbers  $\mathbb{R}$ , and also add in the element  $i$ . We can then take any finite sum of scalar multiples of powers of  $i$ . More formally,

$$\mathbb{R}[i] := \{c_0 + c_1 \cdot i + c_2 \cdot i^2 + c_3 \cdot i^3 + c_4 \cdot i^4 + \cdots + c_n \cdot i^n : n \in \mathbb{N}, c_i \in \mathbb{R}\}.$$

Notice that since  $i^2 = -1 \cdot i^0$ , any power of  $i$  greater than 1 may be re-written as a power less than 2. Thus in practice, we may also write  $\mathbb{R}[i] = \{c_0 + c_1 \cdot i : c_i \in \mathbb{R}\}$ . This shows us why we call  $\mathbb{R}$  a field extension of  $\mathbb{Q}$ —we build the former by literally adding things to the latter.

In general, if  $R$  is a ring,  $R[\alpha]$  is defined similarly.

**Definition 29.** Let  $R$  be an arbitrary ring. If  $\alpha$  is algebraic over  $R$ , then  $R[\alpha] := \{\sum_{i=0}^n c_i \alpha^i : n \in \mathbb{N}, c_i \in R\}$ .

There are some conditions necessary on the value of  $\alpha$ —namely that  $\alpha$  be algebraic. However, algebraic numbers are something we do not address here (for interested readers, this is what transcendental numbers pertain to). Now to make sure things make sense so far, here is a problem.

**Problem 7.2** (Fields; 4, 2, 2, 4).

- a) Examine  $\mathbb{Q}[\sqrt{2}]$ . Setwise, are  $\mathbb{Q}[\sqrt{2}]$  and  $\{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$  equivalent? Why or why not? (Keep in mind the justification of  $\mathbb{R}[i] = \{c_0 + c_1 \cdot i : c_i \in \mathbb{R}\}$  was not fully fleshed out. You must start with an arbitrary maximum degree  $n$  and reduce it to 1.)
- b) Is  $\frac{1}{8}$  inside  $\mathbb{Z}[\frac{1}{2}]$ ? Why or why not?
- c) Is  $\frac{1}{3}$  inside of  $\mathbb{Z}[\frac{1}{6}]$ ? Why or why not?
- d) Are  $\mathbb{Q}[\sqrt{2}]$  and  $\mathbb{Q}[\sqrt{3}]$  equivalent? (Two fields are equivalent if they are setwise equivalent; i.e.  $K$  and  $F$  are equivalent fields if  $K \subset F$  and  $F \subset K$ .) Why or why not?

Finally, a more “professional” way to think about  $\mathbb{R}[i] \cong \mathbb{C}$  comes from realizing that  $i$  is a root of the polynomial  $x^2 + 1$ . Notice that the two roots of  $x^2 + 1$  are  $\pm i$ . This leads to the construction denoted  $\mathbb{R}[x]/(x^2 + 1)$ , which we take to mean that we adjoin to  $\mathbb{R}$  a root of the polynomial  $x^2 + 1$  (it doesn’t matter if we take  $i$  or  $-i$  since they give equivalent fields). In this case, adjoining to  $\mathbb{R}$  a root of  $x^2 + 1$  is exactly the same as adjoining  $i$ . Thus we now have three ways of representing the complex numbers:  $\mathbb{C} \cong \mathbb{R}[i] \cong \mathbb{R}[x]/(x^2 + 1)$ . This latter notation is most important for us. It demonstrates a way of thinking about field extensions: adjoining roots of polynomials. This leads naturally to the concept of Galois groups.

**Definition 30.** For certain rational polynomials  $f(x)$ , the details of which we omit for the sake of time,  $\mathbb{Q}[x]/(f(x))$  is called a Galois extension.

(The specifics of what polynomials are necessary is omitted. They involve definitions which are unnecessary in the scheme of this Power Round, but specific polynomials make their field extensions Galois.) Examine all the roots of  $f(x)$  that exist in this new field  $K$  but don’t exist in  $\mathbb{Q}$ . We can form a group (of functions) that acts on these roots by sending them to each other. For example, for the construction  $\mathbb{Q}[x]/(x^2 + 1)$ , we can imagine a function that sends  $i$  to  $-i$  and vice versa (this is conjugation). The important thing about conjugation is that it sends rational numbers to rational numbers: it fixes elements that

were in the base field. We assert that conjugation and the “do nothing” function (the identity function) are the only such functions that exist. They act on  $\mathbb{Q}[x]/(x^2 + 1)$  but are the identity function when restricted to  $\mathbb{Q}$ . Thus our group of functions has two elements: the conjugation function, and the identity function. We encourage the reader to convince themselves that this small thing is indeed a group. In general, this group that we construct of functions is known as the Galois group of the field extension.

**Definition 31.** Suppose  $K = \mathbb{Q}[x]/(f(x))$  is a Galois extension. Then  $\sigma : K \rightarrow K$  is a Galois automorphism if the following hold:

- If  $a \in \mathbb{Q}$ ,  $\sigma(a) = a$  (this is called “fixing  $a$ ”).
- Let  $A = \{r \in K, r \notin \mathbb{Q} : f(r) = 0\}$ . Then  $\sigma : A \rightarrow A$  is a bijection.
- If  $\alpha, \beta$  are two elements of  $K$ , then  $\sigma(\alpha + \beta) = \sigma(\alpha) + \sigma(\beta)$  and  $\sigma(\alpha \cdot \beta) = \sigma(\alpha) \cdot \sigma(\beta)$ .

**Definition 32.** Let  $G$  be the set of all Galois automorphisms of a field extension  $K$  of  $\mathbb{Q}$ . Then  $G$  is a group that is denoted  $\text{Gal}(K/\mathbb{Q})$  and called the Galois group of  $K$ .

In practice, to form a Galois automorphism, focus first on the second condition. If you impose conditions on where the roots are sent and let rational numbers remain unchanged, the other properties tend to work out as well.

**Problem 7.3** (Galois Automorphisms; 2, 8). Let  $f(x) = x^4 - 70x^2 + 25$ .

- What are the roots of  $f(x)$ ?
- You are given that  $f(x)$  is a nice enough polynomial that  $\mathbb{Q}[x]/(f(x))$  is a Galois. Give four examples of Galois automorphisms—an isomorphism from one set to itself. (You may just tell us where each root is sent for each automorphism).

Before moving on though, we can go even further than what we have done here; we can adjoin multiple roots of many different polynomials at once to  $\mathbb{Q}$ . For example, a classic example you may see if you study mathematics more is adjoining to  $\mathbb{Q}$  the roots of both  $x^2 + x + 1$  and  $x^3 - 2$  at the same time to yield  $\mathbb{Q}[\omega, \sqrt[3]{2}]$  for  $\omega$  a primitive 3rd root of unity.

## 8 Elliptic Curves and Galois Theory

Galois theory is incredibly rich, but unfortunately there are details we must omit about the subject for the sake of time, and this is sufficient background; we can now relate Galois theory and elliptic curves. Suppose you have a general elliptic curve  $E$  and a general point  $P$  on it. We define a  $k$ -division point of  $P$  as some point  $Q$  on  $E$  such that  $kQ = P$ . A fact of elliptic curves is that there are exactly  $k^2$  such  $k$ -division points in  $\mathbb{C}$  (the coordinates of  $Q$  may be complex numbers), but likely many of them won't be rational points. But examine for a moment such a non-rational point  $\beta_k$  such that  $k\beta_k = P$ . Suppose we take the  $x$  and  $y$  coordinates of  $\beta_k$  and adjoined them to  $\mathbb{Q}$ . What would we get? Going further, suppose we took all such  $\beta_{k_i}$  such that  $k\beta_{k_i} = P$  and adjoined to  $\mathbb{Q}$  all of the  $x$  and  $y$  coordinates of these division points. We get some large field extension we label  $K_k$ . This directly gives us a way to use Galois theory in a way that gives us information about our initial  $E$  and  $P$ .

Take on faith that  $K_k/\mathbb{Q}$  is indeed a Galois extension, and examine some Galois automorphism  $\sigma$  of this extension; it acts on all these coordinates we just adjoined. Let  $(a, b)$  be one of the  $k$ -division points; then

$\sigma((a, b)) = (\sigma(a), \sigma(b))$ . Now we have a curious situation: we have found a Galois automorphism that acts on points on an elliptic curve! Weee.

That these Galois automorphisms act on the set of  $k$ -division points is important. Can you visualize how they are acting? These functions send coordinate pairs, essentially vectors, to other vectors. This is quite similar to how matrices act on vectors! In fact, this leads to what is called a Galois representation: a homomorphism from the Galois group to a linear algebra construct. Here we become guilty of omitting some details, but it would take too much work to present in full rigor. But please take these two propositions to be true.

**Proposition 33.** *Let  $E : y^2 + y = x^3 - x$ ,  $P = (0, 0)$ , and let  $K_k$  be the field described above, namely the field extension of  $\mathbb{Q}$  by adjoining all the coordinates of the  $k$ -division points of  $P$ . Then there is a surjective homomorphism from the Galois group  $\text{Gal}(K_k/\mathbb{Q})$  to  $\text{AGL}_2(\mathbb{Z}/2^k\mathbb{Z}) = (\mathbb{Z}/2^k\mathbb{Z})^2 \rtimes \text{GL}_2(\mathbb{Z}/2^k\mathbb{Z})$ . Denote this by  $\varphi : \text{Gal}(K_k/\mathbb{Q}) \rightarrow \text{AGL}_2(\mathbb{Z}/2^k\mathbb{Z})$ .*

A quick word on notation here. We defined the semidirect product in Proposition 16, and here we see an example of one. For an element of such a group  $\text{AGL}_2(\mathbb{Z}/2^k\mathbb{Z})$ , we will write it as  $(\vec{v}, M)$  where  $\vec{v} \in (\mathbb{Z}/2^k\mathbb{Z})^2$  and  $M \in \text{GL}_2(\mathbb{Z}/2^k\mathbb{Z})$ .

**Proposition 34.** *Let  $E : y^2 + y = x^3 - x$  and  $P = (0, 0)$  be a point on  $E$ . Let  $\ell$  be a prime larger than 37. Then  $P$  has odd order in  $E(\mathbb{F}_\ell)$ , the reduction of the curve to this finite field, if and only if for all  $k \in \mathbb{N}$ ,  $\exists(\vec{v}, M) \in \text{AGL}_2(\mathbb{Z}/2^k\mathbb{Z})$  such that  $\vec{v}$  lies in the column space of  $M - I$  (the column space of a matrix such as  $(M - I)$  is the set  $\{(M - I) \cdot \vec{v}, \vec{v} \in (\mathbb{Z}/2^k\mathbb{Z})^2\}$ ).*

Let's parse this last proposition; recall from the interlude that prime  $\ell$  divides some term of the sequence if and only if  $P$  has finite order in  $E(\mathbb{F}_\ell)$ . We saw earlier as well in problem 3.6 that this happens if and only if for all integers  $i$ , there exists some element  $\beta_i \in E(\mathbb{F}_\ell)$  such that  $2^i \cdot \beta_i = P$ . Finally, this condition is equivalent to the latter part of the above proposition. If you are familiar with Galois theory, as a hint of why this might be true, the fact that such a  $\beta_i$  exists implies that it is fixed by the Fröbenius automorphism. This leads to the fact that  $\text{AGL}_2(\mathbb{Z}/2^k\mathbb{Z})$  acting on  $(\mathbb{Z}/2^k\mathbb{Z})^2$  fixes some element  $\vec{x}$ . Thus  $(\vec{v}, M)(\vec{x}) := M \cdot \vec{x} + \vec{v} = \vec{x} \implies (M - I)\vec{x} = -\vec{v}$ . From here it's easy to see that  $\vec{v} \in \text{im}(M - I) \iff -\vec{v} \in \text{im}(M - I)$ .

**Definition 35.** *Let  $(\vec{v}, M) \in \text{AGL}_2(\mathbb{Z}/2^k\mathbb{Z})$  be an element of the affine general linear group. We call  $(\vec{v}, M)$  a ruminative element if  $\vec{v}$  is in the column space of  $M - I$ .*

For the observant reader, another way to describe this element  $(\vec{v}, M)$  is to say that  $M$  fixes a vector  $\vec{x} \in (\mathbb{Z}/2^k\mathbb{Z})^2$  where the action of  $\text{AGL}_2(\mathbb{Z}/2^k\mathbb{Z})$  on  $(\mathbb{Z}/2^k\mathbb{Z})^2$  is as described above.

Thus we have come from an original question about primes dividing terms of a sequence to a question about the column space of matrices. This latter is something that we can much more likely solve directly. (In general, this is a useful strategy. Linear algebra is a subject that is very well understood compared to other mathematical subjects. This is the motivation behind group representations, for example. In fact, that linear algebra is so well understood has given rise to the half-serious joke of dismissing a problem by saying, "it's just linear algebra!") For one final step before we try to use linear algebra to find a fraction, we present the Chebotarev Density Theorem.

**Theorem 36.** *Suppose  $K/\mathbb{Q}$  is a Galois extension with  $G := \text{Gal}(K/\mathbb{Q})$  where  $C \subset G$  is a conjugacy class of  $G$ . Define  $\pi_C(x) := \#\{p \leq x : p \text{ is a prime that is unramified in } K \text{ and } \left[ \frac{K/\mathbb{Q}}{p} \right] = C\}$ . Then*

$$\lim_{x \rightarrow \infty} \frac{\pi_C(x)}{\pi(x)} = \frac{|C|}{|G|}.$$

(The definition of unramified is unimportant for us. In our specific case, this equivalently means primes that are greater than 37.) As written, this doesn't seem to necessarily apply to anything we've written so far. However, one of the facts that we obscured in our presentation of the two propositions above is that the images of the Galois groups above are in fact images of conjugacy classes. The ultimate result of all of this is the following.

**Proposition 37.** *Let  $\pi(x)$  denote the number of primes less than  $x$ , and let  $\pi'(x)$  denote the number of primes less than  $x$  that divide some term of the Tiger sequence. Let  $S$  represent the conjugacy class of  $\text{Gal}(K_k/\mathbb{Q})$  of the  $\beta_k$  such that  $2^k \cdot \beta_k = P$ . Let  $S' := \text{im}(S)$  and  $AGL_2(\mathbb{Z}/2^k\mathbb{Z}) = \text{im}(\text{Gal}(K_k/\mathbb{Q}))$  represent the images under the homomorphism  $\varphi : \text{Gal}(K_k/\mathbb{Q}) \rightarrow AGL_2(\mathbb{Z}/2^k\mathbb{Z})$  from above. Then*

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\pi'(x)}{\pi(x)} &= \lim_{k \rightarrow \infty} \lim_{x \rightarrow \infty} \frac{|S|}{|\text{Gal}(K_k/\mathbb{Q})|} \\ &= \lim_{k \rightarrow \infty} \frac{|S'|}{|AGL_2(\mathbb{Z}/2^k\mathbb{Z})|}. \end{aligned}$$

In other words, the final fraction we have to compute is the last expression in the proposition above. The best way to interpret  $S'$  in the above is that  $S'$  is the subset of  $AGL_2(\mathbb{Z}/2^k\mathbb{Z})$  that consists of the ruminative elements.

## 9 Final Fraction

Thus we are only left with calculating the density! Here are the final steps.

**Definition 38.** Let  $v_p : \mathbb{Z} \rightarrow \mathbb{Z}$  be a function such that  $v_p(n) = m$ , where  $m$  is the exponent of  $p$  in the prime factorization of  $n$ . For example,  $v_2(24) = 3$ ,  $v_3(8) = 0$ , and  $v_5(-25) = 2$ .

**Proposition 39.** Let  $M \in \text{GL}_2(\mathbb{Z}/2^k\mathbb{Z})$  be a matrix such that  $v_2(\det(M - I)) = r$ . Then the number of elements in the column space of  $M - I$  is  $2^{2k-r}$ . (Two notes: we do not regard the cases where  $\det(M - I) = 0$ , and by definition,  $\det(M - I)$  is reduced to be the integer  $n$  such that  $0 \leq n < 2^k$  and  $n \equiv \det(M - I) \pmod{2^k}$  where  $\det(M - I)$  is evaluated in  $\mathbb{Z}$ .)

**Problem 9.1** (Final Fraction; **10, 10, 10, 10**). In all but the last sub-problem here, assume that  $k$  is a fixed positive integer and we examine elements of  $\text{AGL}_2(\mathbb{Z}/2^k\mathbb{Z})$  or  $\text{GL}_2(\mathbb{Z}/2^k\mathbb{Z})$  as the problem dictates. Vectors are arbitrary element  $(\vec{v}, M) \in \text{AGL}_2(\mathbb{Z}/2^k\mathbb{Z})$ .

- a) Error notice: The problem originally stated here was incorrectly phrased. Due to the fact that we are sending out a revision so late, we are awarding everyone the full 10 points for this problem. The problem should have been Proposition 39, which you may assume is true.
- b) Suppose that  $a, b \in \mathbb{Z}/2\mathbb{Z}$ ,  $c \in \mathbb{Z}/2^n\mathbb{Z}$ , and  $n \geq 2$ . Prove the number of pairs  $(\alpha, \beta) \in (\mathbb{Z}/2^n\mathbb{Z})^2$  with  $\alpha\beta \equiv c \pmod{2^n}$  with  $\alpha \equiv a \pmod{2}$  and  $\beta \equiv b \pmod{2}$  is

$$\begin{cases} 0 & ab \not\equiv c \pmod{2}, \\ 2^{n-1} & ab \equiv 0 \pmod{2} \text{ and one of } a \text{ or } b \text{ is nonzero,} \\ (2-1)(v_2(c) - 1)2^{n-1} & a \equiv b \equiv c \equiv 0 \pmod{2}, c \not\equiv 0 \pmod{2^n}, \\ n2^{n-1} & a \equiv b \equiv c \equiv 0 \pmod{2}, c \equiv 0 \pmod{2^n}. \end{cases}$$

(As a note, the factor  $(2 - 1)$  is correctly present, and it is not a typo.)

- c) For  $n \geq 1$ , prove the number of matrices  $M \in \text{GL}_2(\mathbb{Z}/2^k\mathbb{Z})$  with  $\det(M - I) \equiv 0 \pmod{2^{k-1}}$  but with  $\det(M - I) \not\equiv 0 \pmod{2^k}$  is

$$\begin{cases} 2 & k = 1, \\ 3 \cdot 2^{3k-2} - 3 \cdot 2^{2k-1} & k \geq 2. \end{cases}$$

- d) Prove that the density of primes dividing a term of the Tiger sequence is  $\frac{11}{21}$ .

That's it! We hope you've had a fun ride.

# Team Number: \_\_\_\_\_

## PUMaC 2015 Power Round Cover Sheet

Remember that this sheet comes first in your stapled solutions. You should submit solutions for the problems in increasing order. Write on one side of the page only. The start of a solution to a problem should start on a new page. Please mark which questions have been attempted and for which you have submitted solutions to help us keep track of your solutions.

| Problem Number | Points | Attempted? |
|----------------|--------|------------|
| 3.1 a)         | 2      |            |
| 3.1 b)         | 2      |            |
| 3.1 c)         | 2      |            |
| 3.1 d)         | 2      |            |
| 3.2            | 5      |            |
| 3.3            | 5      |            |
| 3.4            | 5      |            |
| 3.5            | 10     |            |
| 3.6            | 5      |            |
| 4.1 a)         | 2      |            |
| 4.1 b)         | 8      |            |
| 4.2 a)         | 2      |            |
| 4.2 b)         | 2      |            |
| 4.3 a)         | 2      |            |
| 4.3 b)         | 8      |            |
| 4.4            | 5      |            |
| 4.5 a)         | 2      |            |
| 4.5 b)         | 2      |            |
| 4.6            | 10     |            |

| Problem Number | Points | Attempted? |
|----------------|--------|------------|
| 5.1 a)         | 5      |            |
| 5.1 b)         | 2      |            |
| 5.2            | 10     |            |
| 5.3            | 5      |            |
| 5.4 a)         | 8      |            |
| 5.4 b)         | 12     |            |
| 6.1            | 5      |            |
| 6.2            | 5      |            |
| 7.1            | 5      |            |
| 7.2 a)         | 4      |            |
| 7.2 b)         | 2      |            |
| 7.2 c)         | 2      |            |
| 7.2 d)         | 4      |            |
| 7.3 a)         | 2      |            |
| 7.3 b)         | 8      |            |
| 9.1 a)         | 10     | ✓          |
| 9.1 b)         | 10     |            |
| 9.1 c)         | 10     |            |
| 9.1 d)         | 10     |            |
| Total:         | 200    | $\geq 10$  |

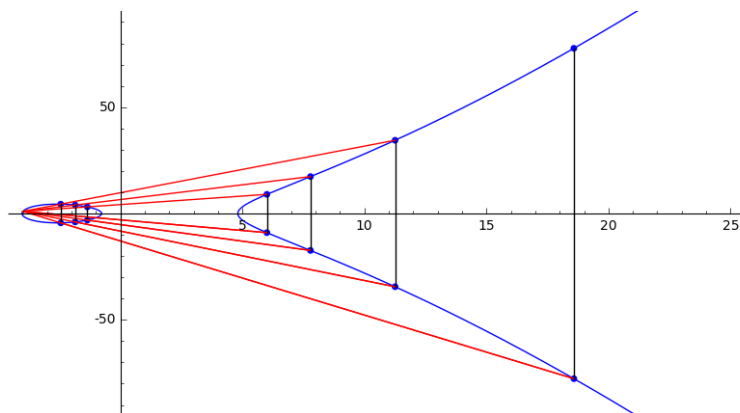


Figure 4: Graphical View of kP.